Chapter 10: Rotational Motion

The following subjects will be discussed

- Rotational/angular Motion Variables.
- Equations of Motion.
- Inertia of an Object.
- Kinetic Energy.
- Torque and Newton's Laws.
- Work in Angular Motion.

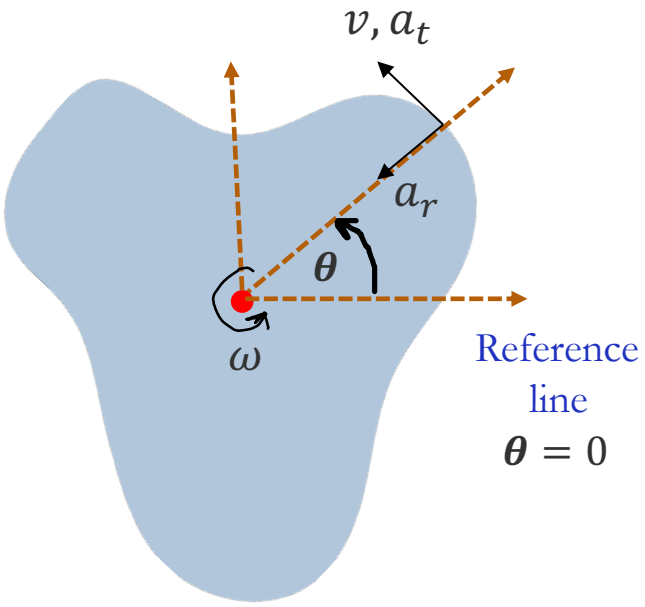
Equations of Angular Motion

For constant acceleration

$$\omega(t) = \omega_o + \alpha t$$

$$\theta(t) = \theta_o + \omega_o t + \frac{1}{2}\alpha t^2$$

$$\omega^2(t) = {\omega_o}^2 + 2\alpha \cdot [\theta(t) - \theta_o]$$



Angular-linear relations

$$S = \theta \cdot r$$

$$v = \omega \cdot r$$

$$a_t = \alpha \cdot r \qquad a_r = \omega^2 \cdot r$$

Examples

1) At t=0, the flywheel has an angular velocity of 4.7 rad/s and a constant angular acceleration of -0.25 rad/s^2. What is the maximum angle does the wheel go through before it stops? (reference is at t=0)

$$\omega^2(t) = {\omega_o}^2 + 2\alpha \cdot [\theta(t) - \theta_o]$$

$$\theta_{max} = \frac{\omega^2 - {\omega_o}^2}{2\alpha} = \frac{0^2 - 4.7^2}{-2(0.25)} = 44.18\, rad$$



How much time?

$$\omega(t) = \omega_o + \alpha t$$

$$0 = 4.7 + (-0.25)t$$

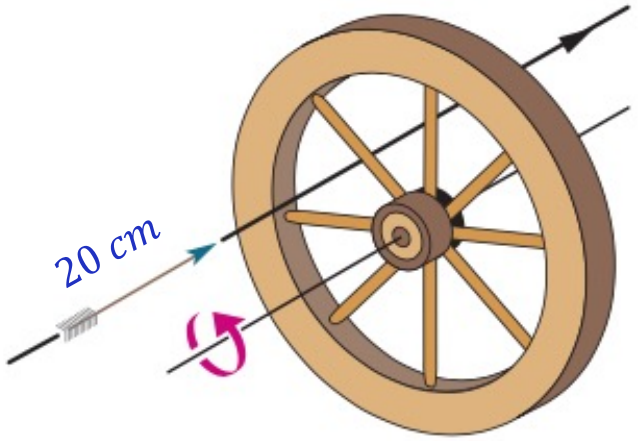
$$t = 18.8\, s$$

What happens after it stops?

Examples

2) The wheel has 8 equal slots. For the arrow to make a clean pass through, what should be its minimum speed?

$$\omega = 2.5\ \frac{rev}{s} \cdot \frac{2\pi\ rad}{1\ rev} = 5\pi\ \ rad/s$$



$r = 30\ cm$

$\omega = 2.5\ rev/s$

Time needed to pass = time between slots,

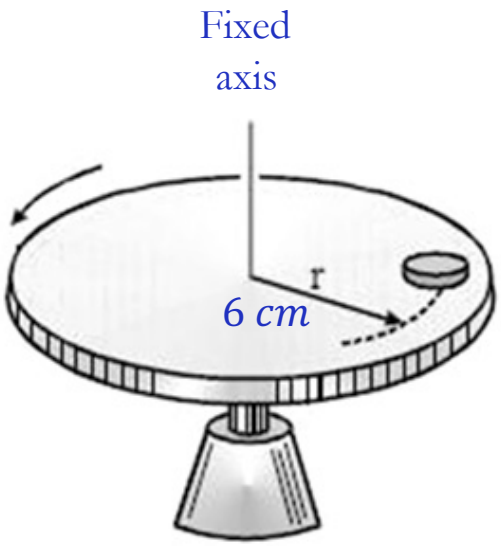
$$t = \frac{l_{arrow}}{v_{arrow}} = \frac{T}{8} = \frac{2\pi}{8\omega} = \frac{2\pi}{8(5\pi)}$$

$$v_{arrow} = (0.2)(20) = 4\ m/s$$

Where should the arrow be aimed?

Examples

3) The turning table is a rough surface and rotating at a constant speed. a) Find the radial acceleration of the coin. 2) the required static coefficient of friction between the coin and the table to prevent slipping.



$$\omega = 33\frac{1}{3}\frac{rev}{min} = 3.5\frac{\text{rad}}{\text{s}}$$

a) the radial acceleration $a_r = \omega^2 \cdot r$

$$a_r = (3.5)^2(6) = 73.3\ cm/s^2$$

b) static coefficient of friction

$$ma_r = \mu_s mg \qquad \mu_s = \frac{a_r}{9.81} = 0.075$$

c) Repeat (b) if the table has constant angular acceleration of 14 rad/s^2.

$$a_t = \alpha \cdot r$$

$$a_t = (14)(0.06) = 0.84\ m/s^2$$

$$a = \sqrt{{a_t}^2 + {a_r}^2} = \sqrt{0.84^2 + 0.733^2} = 1.115\ \ m/s^2$$

$$\mu_s = \frac{a}{9.81} = 0.114$$

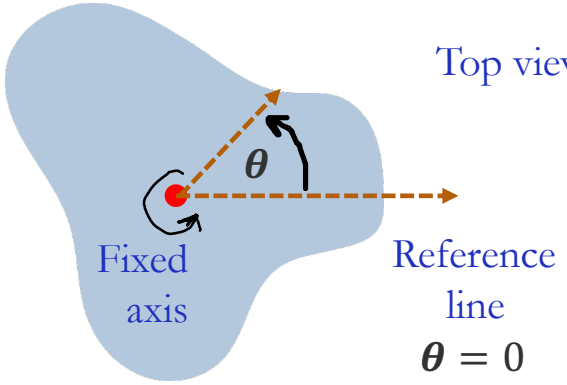
Kinetic Energy

The formula for kinetic energy for linear (translational) motion ($KE = \frac{1}{2}mv^2$) is not going to work here for two reasons. 1) not all parts of the object move with the same linear speed and 2) the speed of the center of mass is zero.

We need to consider individual particles of the object, as

$$KE = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \cdots + \frac{1}{2}m_nv_n^2$$

$$KE = \sum_{i=1}^{n}\frac{1}{2}m_iv_i^2$$



With $v = \omega \cdot r$

$$KE = \frac{1}{2}\sum_{i=1}^{n}m_ir_i^2 \cdot \omega^2 \quad \rightarrow \quad KE = \frac{1}{2}\left(\sum_{i=1}^{n}m_ir_i^2\right)\omega^2$$

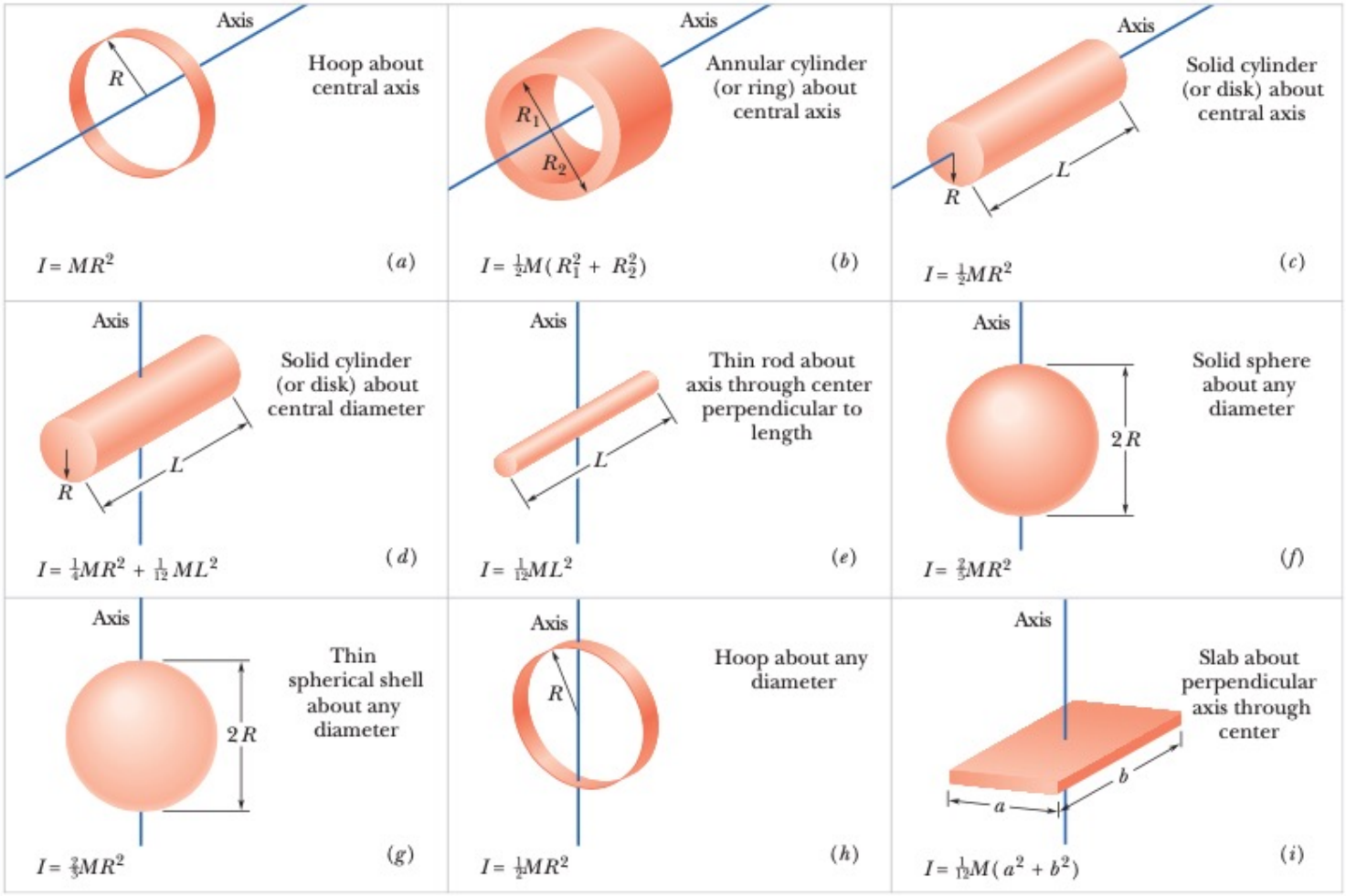
I is termed the "inertia" of an object with units of kg m^2

$$I = \sum_{i=1}^{n}m_ir_i^2 \qquad I = \int r^2\, dm$$

$$\rightarrow \quad KE = \frac{1}{2}\, I\, \omega^2$$

Inertia Calculations

For some symmetric (and nice) shapes, we have "closed form" relations for the inertia.



Inertia Calculations

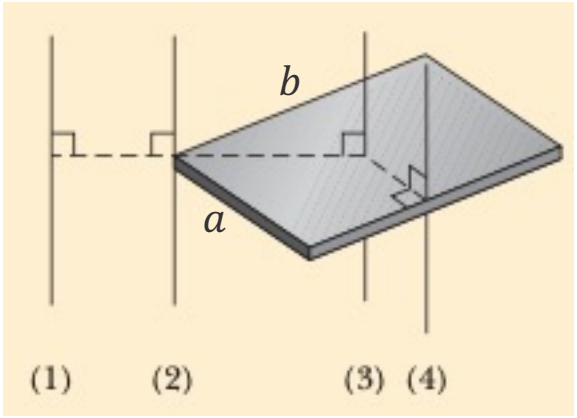
For different rotation axes, can also utilize the "parallel-axis" theorem

$$I = I_{cm} + Mh^2$$

$$I_3 = I_{cm} = \frac{1}{12} M(a^2 + b^2)$$

$$I_4 = I_{cm} + M \frac{a^2}{4}$$

$$I_2 = I_{cm} + M(\frac{a^2}{4} + \frac{b^2}{4})$$

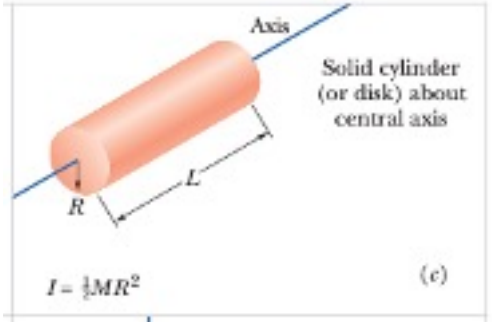


Examples

1) A solid cylinder of mass 500 kg and radius 1 m is rotated at its central axis at 200π rad/s. Find its kinetic energy.

$$I = \frac{1}{2}MR^2$$

$$I = \frac{1}{2}(500)(1)^2 = 250 \ \ kg \cdot m^2$$



$$KE = \frac{1}{2}\, I\, \omega^2$$

$$KE = \frac{1}{2}\,(250)(200\pi)^2 = 49.35 \ \ MJ$$

In how much time would this energy be consumed at an average power of 8 kW?

$$P_{ave} = \frac{energy}{time}$$

$$t = \frac{49.35 \ \ MJ}{8 \ kW} = 102.8 \ min$$

Examples

2) For the arrangement below, find the system inertia and kinetic energy.

$$I_{rod_1} = \frac{1}{12}(1.2)(0.056)^2 + (1.2)\left(\frac{0.056}{2}\right)^2$$

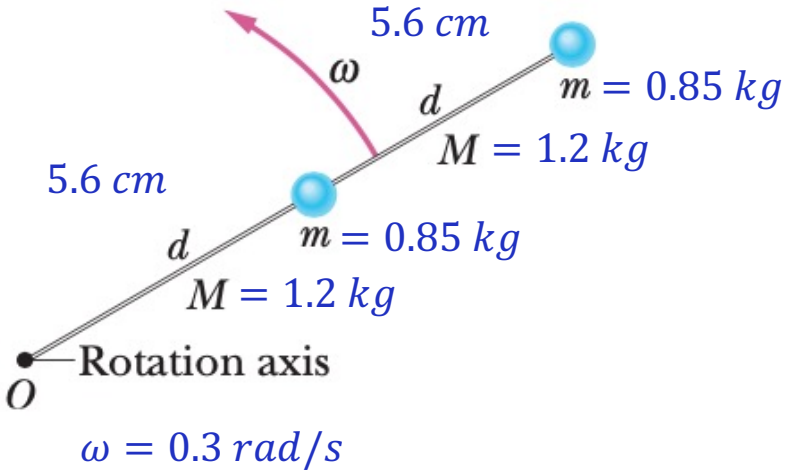
$$I_{rod_2} = \frac{1}{12}(1.2)(0.056)^2 + (1.2)\left(0.056 + \frac{0.056}{2}\right)^2$$

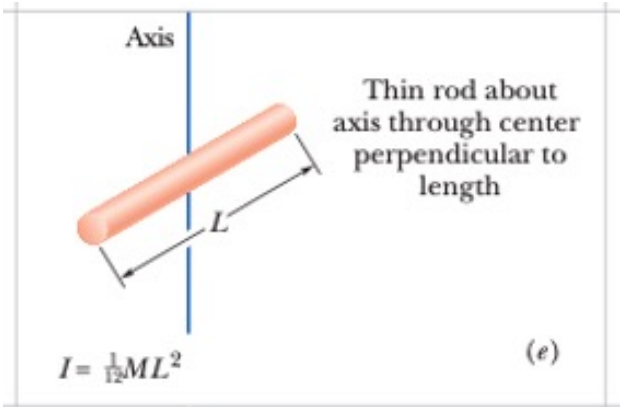
$$I_{m_1} = (0.85)(0.056)^2$$

$$I_{m_2} = (0.85)(2\times 0.056)^2$$

$$I_{tot} = I_{m_1} + I_{m_2} + I_{rod_1} + I_{rod_2} = 0.0233 \ \ kg \cdot m^2$$

$$KE = \frac{1}{2}(0.0233)(0.3)^2 = 1.1 \ mJ$$





Newton's Second Law for Rotation

Here, an external force is applied to cause rotation. However, only the tangential component of the applied force would cause rotation.

$$F_t = F \sin\theta$$

We need to relate this force to the rotational motion variables.

$$F_t = m\, a_t$$

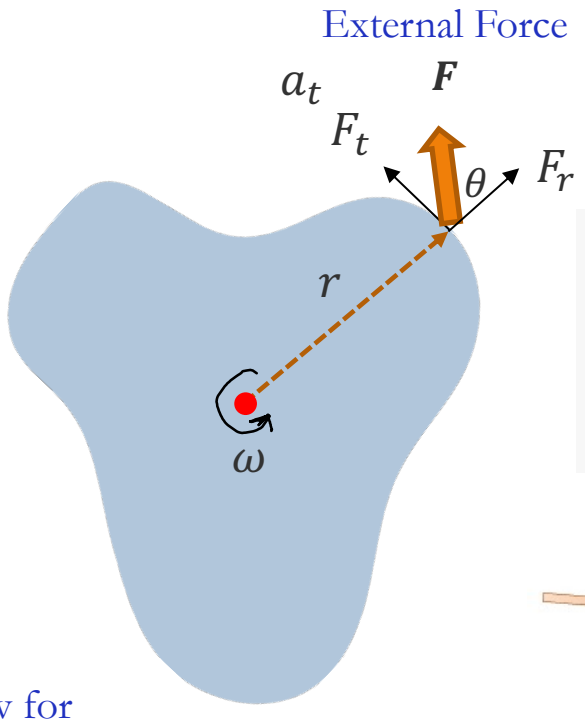
$$F_t = m\, \alpha\, r$$

$$F_t\, r = (mr^2)\, \alpha$$

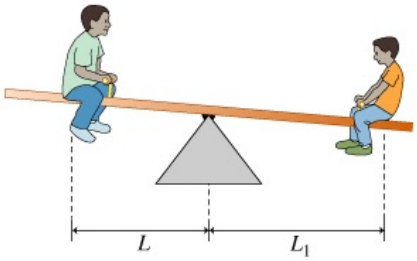
We define "Torque" → $\tau = I\, \alpha$

Units N.m

Newton's Second Law for Rotational motion







$$\tau_{net} = F_{t_{net}}\, r$$

Positive torque, counter-clockwise (CCW) motion.

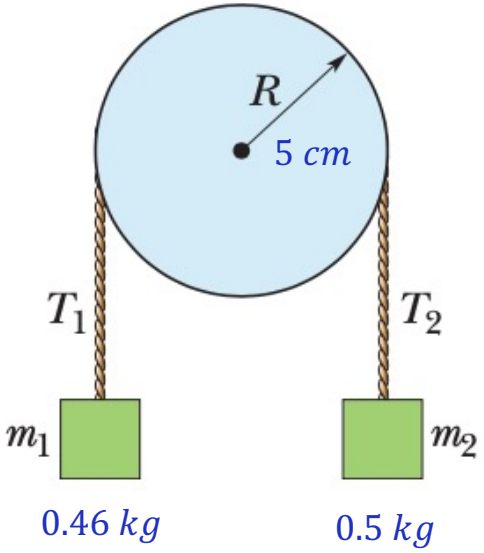
Negative torque, clockwise (CW) motion.

Examples

1) The system below is initially at rest. At t=0, mass 2 falls 75 cm in 5 s, without the rope slipping (pulley rotates).

Find the inertia of the pulley.

We can find the inertia from Newton's law $\tau = I\,\alpha$, but we need the angular acceleration and torque.



a) Find linear acceleration

$$y - y_o = v_o t + \frac{1}{2}at^2 \quad \rightarrow \quad 0.75 = \frac{1}{2}a(5)^2 \quad \rightarrow \quad a = 0.06 \text{ m/s}^2$$

b) Find angular acceleration $\alpha = \frac{a_t}{r} = \frac{0.06}{0.05} = 1.2 \text{ rad/s}^2$

c) Find torque $\tau_{net} = F_{t_{net}}\, r = (T_2 - T_1)\, R$

$$\tau_{net} = 0.33\ (0.05)$$

$$I = \frac{0.33(0.05)}{1.2} = 0.0138\ kg \cdot m$$

$$m_2 g - T_2 = m_2 a$$

$$T_1 - m_1 g = m_1 a$$

$$(T_1 - T_2) + (m_2 - m_1)g = +(m_2 + m_1)a$$

$$(T_2 - T_1) = 0.33\ N$$

Q & A

Send me an email if
you have any
questions.